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Quantum meaningful visual cryptography

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Abstract

The quantum visual encryption scheme utilizes quantum property to encrypt secret information, which increases the security of secret information. However, its share images are meaningless noisy images. Therefore, the share image is vulnerable to be attacked after quantum measurement. To further improve the security of image data transmission, we propose a (n, n) quantum meaningful visual encryption scheme. The scheme consists of two phases, the encryption phase and the recovery phase. The secret image is a binary image and n cover images are halftone images. In the encryption phase, either the secret image pixel or the cover image pixel is randomly selected for encryption by a quantum selection circuit. The color information of each pixel of the secret image and the cover image is then encoded as n-qubit superposition state. Finally, these qubits are divided among the n participants. In the recovery stage, n participants work together to collect all the qubits of each pixel and perform n-qubit OR operation to recover the secret information. Simulations were carried out to verify the feasibility of this scheme. The share images in this scheme are meaningful and retain the advantage of single-pixel parallel processing.

Keywords Meaningful share image · Quantum visual cryptography · Single-pixel parallel processing · n-qubit OR operation

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1 Introduction

In recent years, with the rapid development of modern science and technology, data transmission is filled in every aspect of our life. Image as an important carrier of data, its security cannot be ignored. Therefore, how to ensure the security of data transmission has become a research hotspot. Visual cryptography (VC) [1], as one of the important research directions in the field of information security, is simple in the encryption process and requires no computation in the decoding process. It can recover secret image through the human visual system (HVS), which has received extensive attention.

Naor and Shamir proposed the VC scheme for the first time. In the encryption phase, the design matrix is adopted to encrypt the secret image using pixel extension. In the decryption phase, the secret image can be recovered by directly stacking the share image. However, the share image is similar to noise, which is easy to attract the attention of attackers. Therefore, Naor proposed a simple (2,2) meaningful visual cryptography scheme in his paper. The share image displays the color information of the cover image and the secret image is recovered after stacking. However, both the cover image and the recovered secret image have poor-quality problems due to the existence of pixel expansion. Most of the meaningful visual secret schemes proposed later focus on reducing pixel expansion and improving the quality of secret image recovery. Kafri [2] proposed a random grid visual encryption scheme (RG-VC), where one share is generated randomly and the other share is determined according to the color of the secret message. This scheme has no pixel expansion, but its decryption number is limited. After that, (n, n) threshold RG-VC [3], (2,n) threshold RG-VC [4, 5] , and (t, n) threshold RG-VC [6, 7] were proposed successively, which were no longer limited by the number of participants. However, the share images of the above schemes were meaningless noisy images. Zhou [8] proposed a half-tone visual cryptography scheme, which used the method of error diffusion to embed the cover image into the share image. Meanwhile, the secret pixel used the traditional visual encryption scheme to generate sub-pixels and embed them into the share images to achieve meaningful visual cryptography, but there was a problem with pixel expansion. Chen [9] proposed a user-friendly visual encryption scheme based on the random grid. The secret image is encrypted by the encryption method of random grid, and the appearance of the share image displays cover image. After stacking, the secret image is displayed, and the cover image will become noise without affecting the recognition of secret information. However, it is a (2,2) threshold encryption scheme, and the number of participants is limited. Yan [10] extends it to the general access structure and is no longer limited to two shares.

With the rapid development of quantum computer, its advantages have been known to a growing number of people. A large number of quantum algorithms have been proposed, such as Shor's algorithm and Grover's algorithm. They enable traditional cryptography to be broken in a very short period of time and pose a huge threat to traditional cryptography schemes. The security of classical cryptography schemes is reduced, no longer meeting the needs of information security. Therefore, many researchers extend the classical scheme to the quantum field, hoping to improve the performance of the classical cryptography scheme by using quantum properties, such

as quantum key distribution [11–14], quantum digital signature [15], quantum secure direct communication [16], and quantum conference key agreement [17]. Hillery [18] proposed the first quantum secret sharing (QSS) scheme. Quantum-based secret sharing schemes have been proposed continuously. Their schemes are diverse and can be divided into three categories: QSS of classical information [19–23], QSS of quantum information [24–29], and QSS of combining classical information with quantum information [30]. Inspired by it, the researchers combined quantum and visual cryptography to propose the scheme of quantum visual cryptography.

Tseng and Hwang [31] first combined quantum signal processing and VC schemes; however, their scheme did not overcome the drawbacks of traditional VC schemes. Later, Das [32] proposed a visual encryption scheme based on quantum signal processing, which normalized the gray value of the image pixel in the range of 0 and 1. The input pixel is converted into a qubit. The ground states of the qubit is all possible encryption methods for the pixel. The quantum measurement probability is used to determine which encryption method to choose. The share image is generated according to the intensity of the input pixel. It applies to all image types without pixel expansion. Liu [33] proposed a novel (n, n) quantum visual secret sharing scheme (QVSS), which encodes the color information of secret image pixel into n-qubit superposition state. Assign n qubits to n participants; each participant holds one qubit. This scheme is decrypted by quantum XOR operation. Later, Liu [34] proposed a (t, n) threshold QVSS scheme, which encodes the color information of secret image pixel into $n \times m$-qubit superposition state and distributes them to n participants. Each participant holds m qubits. In the decryption phase, at least t participants are in the process of decryption. The color information of secret pixel is determined by Hamming weights. Zhu [35] proposed a quantum progressive visual secret sharing scheme (QPVSS). The color information of the secret pixel is encoded as a quantum superposition state through matrix. The decryption phase is completed by two XOR operations. As the number of participants increases, the quality of the recovered image is improved. However, the above schemes remain meaningless images after quantum measurement.

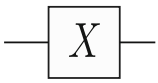
In order to further improve the security of secret information, we propose a (n, n) quantum meaningful visual cryptography (QMVC) scheme. The share images are meaningful images that can divert the attacker's attention after the quantum measurement, further securing the secret images.

This paper is organized as follows. In Sect. 2, we briefly describe the quantum gates required to design the quantum circuits in the scheme of this paper. In Sect. 3, our proposed quantum meaningful visual encryption scheme is presented. Section 4 conducts simulated quantum experiments. The results of the experiments are analyzed and discussed in Sect. 5. Finally, conclusions are given in Sect. 6.

2 Preliminaries

In this section, we introduce the basic knowledge of quantum computing related to this paper. The qubit is introduced in Sect. 2.1. In Sect. 2.2, quantum logic gates are introduced.

Fig. 1 Quantum X gate



2.1 Qubit

The basic unit of classical information is the bit, which has two states of 0 and 1. Similar to classical bits, the qubit is the basic unit of quantum information. In addition to having two definite states $|0\rangle$ and $|1\rangle$, a qubit can also be in the case of a superposition of two states, which is usually called a quantum superposition state,

$$|q\rangle = a\,|0\rangle + b\,|1\rangle\,, \tag{1}$$

where a and b are called probability amplitude, $a, b \in \mathbb{C}$, $\mathbb{C}$ is complex number, and satisfy $|a|^2 + |b|^2 = 1$. After the quantum measurement, $|q\rangle$ collapses into a certain quantum state. The probability of collapsing into $|0\rangle$ is $|a|^2$, and the probability of collapsing into $|1\rangle$ is $|b|^2$. The vector representation of $|0\rangle$ and $|1\rangle$ is shown as:

$$|0\rangle = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, |1\rangle = \begin{bmatrix} 0 \\ 1 \end{bmatrix}. \tag{2}$$

2.2 Quantum gate

The quantum logic gate is the basic component of the quantum circuit. The quantum state will change from one state to another after passing through the quantum gate. Therefore, the role of the quantum gate is to change the quantum state. Based on the linear property, quantum gates can be expressed in the form of the matrix. The quantum gates used in this paper are shown below.

The X-gate is to take the inverse of the quantum state. The transformation result is $|q\rangle = a\,|1\rangle + b\,|0\rangle$ when the X-gate acts on the quantum state $|q\rangle = a\,|0\rangle + b\,|1\rangle$. The quantum X-gate is shown in Fig. 1. The X-gate can be represented by the Pauli-X matrix:

$$\mathrm{X} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}. \tag{3}$$

The Hadamard gate enables the quantum state to be in a superposition state of $|0\rangle$ and $|1\rangle$. The Hadamard gate is referred to as the H-gate. The quantum H-gate is shown in Fig. 2. The matrix is represented as:

$$\mathrm{H} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}. \tag{4}$$

Fig. 2 Quantum H gate

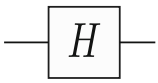


Fig. 3 Quantum Y rotation gate

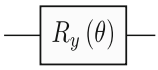
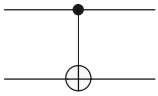


Fig. 4 Quantum CNOT gate



When the H-gate is applied to the ground state $|0\rangle$ or $|1\rangle$, the result can be expressed as:

$$H|0\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), H|1\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle). \tag{5}$$

The quantum Y rotation gate can change the probability of quantum superposition state collapsing into different ground states by adjusting the angle. The quantum Y rotation gate is shown in Fig. 3. The matrix is represented as:

$$\mathrm{R_y}(\theta) = \begin{bmatrix} \cos\frac{\theta}{2} & -\sin\frac{\theta}{2} \\ \sin\frac{\theta}{2} & \cos\frac{\theta}{2} \end{bmatrix}. \tag{6}$$

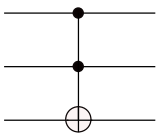
The CNOT gate is a two-qubit gate, one of which is the control qubit and the other is the target qubit. When the control qubit is $|1\rangle$, the target qubit flips. When the control qubit is $|0\rangle$, the target qubit remains unchanged. The CNOT gate is shown in Fig. 4, and its matrix is denoted as:

$$\mathrm{CNOT} = \begin{bmatrix} 1&0&0&0 \\ 0&1&0&0 \\ 0&0&0&1 \\ 0&0&1&0 \end{bmatrix}. \tag{7}$$

The Toffoli gate is a three-qubit gate, two of which are control qubits, and the remaining one is target qubit. The target qubit flips when both control qubits are $|1\rangle$, while in other cases the target qubit remains unchanged. The Toffoli gate is shown in Fig. 5, and its matrix is denoted as:

$$\mathrm{T} = \begin{bmatrix} 1&0&0&0&0&0&0&0 \\ 0&1&0&0&0&0&0&0 \\ 0&0&1&0&0&0&0&0 \\ 0&0&0&1&0&0&0&0 \\ 0&0&0&0&1&0&0&0 \\ 0&0&0&0&0&0&1&0 \\ 0&0&0&0&0&1&0&0 \\ 0&0&0&0&0&0&0&1 \end{bmatrix}. \tag{8}$$

Fig. 5 Quantum Toffoli gate



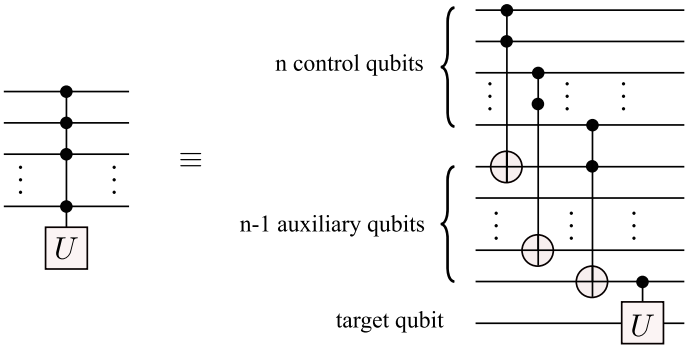


Fig. 6 Quantum $C^n - U$ gate

A multiply controlled U gate ($C^n - U$) has multiple control qubits. When the control qubits are all $|1\rangle$, the corresponding U operation acts on the target qubit; otherwise, the U gate does not work. The quantum circuit is shown in Fig. 6, where U is an arbitrary single-bit quantum gate. In this paper, $C^n - X$ and $C^n - R_y(\theta)$ are used. A $C^n - U$ gate consists of multiple Toffoli gates and a controlled U gate (U_{CU}). The matrix of U_{CU} is expressed as:

$$\mathrm{U_{CU}} = \begin{bmatrix} \mathbb{1} & 0 \\ 0 & U \end{bmatrix}, \tag{9}$$

where $\mathbb{1}$ is the Identity matrix.

3 Proposed scheme

In this section, we propose a quantum meaningful visual cryptography scheme based on the scheme of Chen et al. [9]. The secret image is a binary image, and the cover images are halftone images. In Sect. 3.1, we give the design of the overall scheme. In Sect. 3.2, we give the detailed steps of the encryption phase. In Sect. 3.3, the decryption process is described.

3.1 Overall scheme design

The distributor wants to send a secret image of size $M \times N$ to n participants $\{P_1, P_2, \ldots, P_n\}$, requiring a total of n cover images $\left\{C^1, C^2, \cdots, C^n\right\}$. The overall block diagram of a quantum meaningful visual cryptography scheme is shown in Fig. 7.

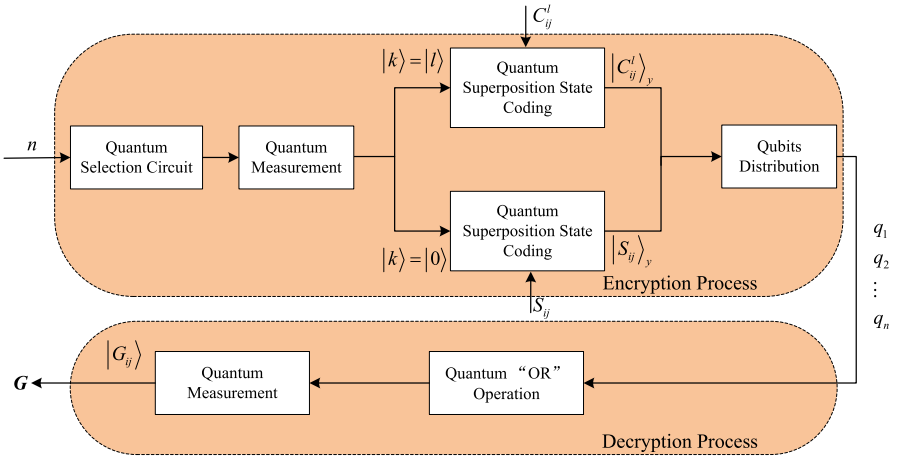


Fig. 7 The overall block diagram of quantum meaningful visual encryption scheme

In the encryption phase, the quantum selection circuit is first designed according to the number of participants n. The number of qubits of the quantum selection circuit is $t\left(t = \lceil \log_2^{(n+1)} \rceil\right)$. The quantum measurement of the quantum selection circuit was performed to obtain the output $|k\rangle$. The value of $|k\rangle$ determines whether to encrypt the color information of the secret image pixel S_{ij} ($i \in \{1, 2, \ldots, M\}$, $j \in \{1, 2, \ldots, N\}$) or the information of the cover image pixel C_{ij}^{l} ($l \in \{1, 2, \ldots, n\}$). If the measurement result of the quantum selection circuit is considered as a binary representation, then k is the decimal representation of the result. For example, if $|k\rangle = |100\rangle$, then $k = 4$. If $|k\rangle = |0\rangle$, it means encrypting S_{ij}. The S_{ij} is encoded as an n-qubit superposition state $\left|S_{ij}\right\rangle_y$, where $y \in \{0, 1\}$, the color of S_{ij} is white when $y = 0$, and the color of S_{ij} is black when $y = 1$. If $|k\rangle = |l\rangle$, it means that the l-th cover image pixel information C_{ij}^{l} is encrypted. Encode C_{ij}^{l} into the n-qubit superposition state $\left|C_{ij}^{l}\right\rangle_y$. Finally, the encoded n qubits$\{q_1, q_2, \ldots, q_n\}$ are sent to the n participants.

In the decryption phase, n participants work together to recover the secret image. The recovered image $\boldsymbol{G}$ is obtained by performing OR operation on n qubits of a pixel.

3.2 Encryption phase

The encryption phase consists of three parts: the design of the quantum selection circuit, the encoding of secret image pixel information and cover image pixel information, and the allocation of qubits.

The quantum selection circuit is first designed based on n. As shown in Fig. 8, one rotation Y gate and $n - 1$ controlled rotation Y gates are required. ∘ means the control bit is $|0\rangle$, and • means the control bit is $|1\rangle$. The circuit is in a superposition of $n + 1$ ground states. By adjusting the rotation angle of the quantum rotation Y gate and the controlled quantum rotation Y gate, we can change the probability of the quantum selection circuit collapsing to different ground states. For example, when $n = 2$, the

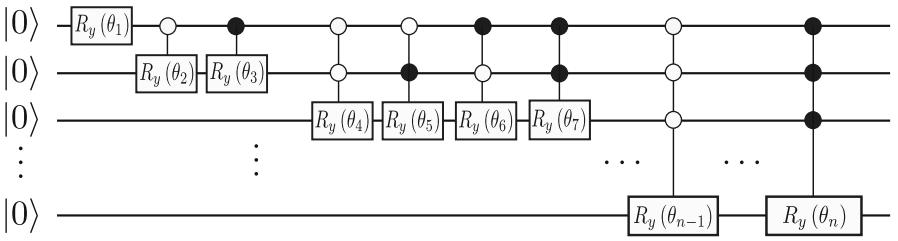


Fig. 8 The quantum selection circuit

quantum selection circuit requires one rotation Y gate and one controlled rotation Y gate. The quantum selection circuit is in a superposition of three ground states of two qubits. The result is expressed as:

$$|k\rangle = \cos\frac{\theta_1}{2}\cos\frac{\theta_2}{2}|00\rangle + \cos\frac{\theta_1}{2}\sin\frac{\theta_2}{2}|01\rangle + \sin\frac{\theta_1}{2}|10\rangle. \tag{10}$$

After the quantum measurement, the quantum superposition state $|k\rangle$ collapses to a definite quantum state. The probability of collapse is the squared value of the corresponding amplitude.

Then, we encode the secret image pixel S_{ij} or cover image pixel C_{ij}^l as quantum superposition state.

For S_{ij}, we encode the quantum superposition state according to the encryption of the (n, n) random grid. The form of the quantum superposition state can be expressed as:

$$\left|S_{ij}\right\rangle_y = \frac{1}{\sqrt{2^{n-1}}}\sum_{h=1}^{2^{n-1}}\left|m_h^1 m_h^2 \cdots m_h^n\right\rangle, \tag{11}$$

where $y \in \{0, 1\}, m_h \in \{0, 1\}$. If the color of S_{ij} is white, the quantum circuit is shown in Fig. 9a. The circuit is designed using $n-1$ H gates and $n-1$ CNOT gates to control the circuit. The circuit has a total of 2^{n-1} ground states, and the number of $|1\rangle$ in each ground state is even. There is only one ground state of the 2^{n-1} ground states of $\left|S_{ij}\right\rangle_0$ in which all the qubits are $|0\rangle$, so the probability of getting $|0\rangle$ after decoding is $\frac{1}{2^{n-1}}$. For example, when $n = 4$, the quantum superposition state $\left|S_{ij}\right\rangle_0$ has a total of 8 ground states $\{|0000\rangle, |0011\rangle, |0101\rangle, |0110\rangle, |1001\rangle, |1010\rangle, |1100\rangle, |1111\rangle\}$. The result of the ground state $|0000\rangle$ after the quantum OR operation is $|0\rangle$, and the result of all other ground states after the quantum OR operation is $|1\rangle$. Therefore, the probability of getting $|0\rangle$ after decoding is $\frac{1}{8}$. If the color of S_{ij} is black, the quantum circuit is shown in Fig. 9b. After adding an X gate at the n-th qubit, the number of $|1\rangle$ in the quantum superposition state $\left|S_{ij}\right\rangle_1$ is odd. The result of the decoding of the 2^{n-1} ground states of the quantum superposition state $\left|S_{ij}\right\rangle_1$ is all $|1\rangle$.

For C_{ij}^l, we encode the quantum superposition state based on the color information of C_{ij}^l. The quantum circuit is shown in Fig. 10, where n quantum Y rotation gates

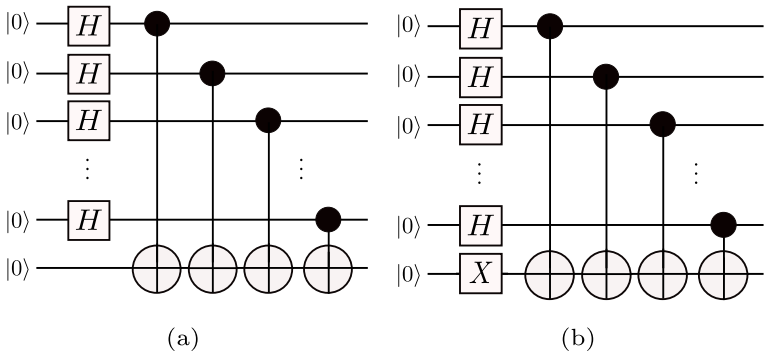


Fig. 9 **a** The quantum encryption circuit for white pixel **b** The quantum encryption circuit for black pixel

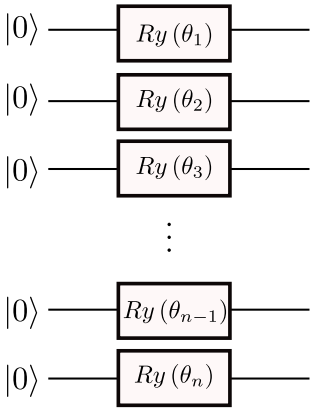


Fig. 10 The quantum state encoding circuit for cover images

are used to change the probability of their collapse into different ground states. If C_{ij}^{l} is white, the rotation angle θ_l is adjusted to encode the l-th position of the quantum circuit with a high probability of $|0\rangle$. If C_{ij}^{l} is black, the rotation angle θ_l is adjusted to encode the l-th position of the quantum circuit with a high probability of $|1\rangle$. In order to ensure that this position does not affect the secret information after decoding, the pixel of the remaining cover images $\left\{C_{ij}^{1}, C_{ij}^{2}, \ldots, C_{ij}^{l-1}, C_{ij}^{l+1}, \ldots, C_{ij}^{n}\right\}$ at this position are encoded as a qubit with a high probability of $|1\rangle$. The form of the quantum superposition state can be expressed as:

$$\left|C_{ij}^{l}\right\rangle_{y} = \bigotimes_{h=1}^{n} \cos\left(\theta_h\right)|0\rangle + \sin\left(\theta_h\right)|1\rangle, \tag{12}$$

where $\otimes$ is the tensor product and $\left|C_{ij}^{l}\right\rangle$ has 2^n ground states.

Then, the distributor distributes n qubits$\{q_1, q_2, \ldots, q_n\}$ to the n participants $\{P_1, P_2, \ldots, P_n\}$. Each participant received one qubit, so they could not get information about the secret image. Meanwhile, the total qubits matrix obtained by each

participant is a cover image. The encryption process of QMVC can be summarized in Algorithm 1.

Algorithm 1 QMVC encryption algorithm.

Require: secret image S, n cover images C^l ($l \in (1, 2, \ldots, n)$), the number of participants n.
Ensure: n shares of qubits.

```
for i = 1 to M do
  for j = 1 to N do
    Construction of quantum selective circuit.
    Quantum measurement performed on quantum selection circuit to obtain |k⟩.
    if |k⟩ = |0⟩ then
      if the color of S_ij is white. then
        S_ij → |S_ij⟩_0
      else
        S_ij → |S_ij⟩_1
      end if
    else if |k⟩ = |l⟩ then
      if the color of C^l_ij is white. then
        C^l_ij → |C^l_ij⟩_0
      else
        C^l_ij → |C^l_ij⟩_1
      end if
    end if
    for b = 1 to n do
      P_b ← q_b
    end for
  end for
end for
```

3.3 Decryption phase

Suppose P_r ($1 \leq r \leq n$) is assigned to recover the secret image. He first collects all the qubits$\{q_1, q_2, \ldots, q_n\}$ of a pixel in the n participants $\{P_1, P_2, \ldots, P_n\}$. The n qubits are then passed through a quantum OR operation circuit, and the quantum circuit is shown in Fig. 11. When all n qubits are $|0\rangle$, the result of quantum measurement is $|0\rangle$. If any one of the n qubits is not $|0\rangle$, the result of quantum measurement will always be $|1\rangle$. If the quantum measurement result is $|0\rangle$, the recovered image pixel G_{ij} is white; if the quantum measurement result is $|1\rangle$, the recovered image pixel G_{ij} is black. The decryption process for QMVC is shown in Algorithm 2.

4 Experiment implementation

In this section, we perform experimental verification of the algorithm proposed in Sect. 3 using Qiskit. Qiskit [quiss-kit] is an open-source SDK for working with quantum computers at the level of pulses, circuits, and application modules. In Sect. 4.1,

Algorithm 2 QMVC decryption algorithm.

Require: n qubits.
Ensure: Recovered image $\boldsymbol{G}$.
1: **for** $i = 1$ to M **do**
2: **for** $j = 1$ to N **do**
3: $q_1 q_2 \cdots q_n \rightarrow \left|G_{ij}\right\rangle$
4: **if** $\left|G_{ij}\right\rangle = |0\rangle$ **then**
5: the color of G_{ij} is white.
6: **else**
7: the color of G_{ij} is black.
8: **end if**
9: **end for**
10: **end for**

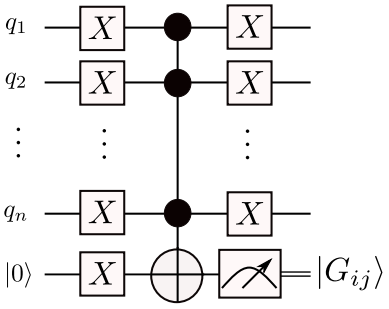


Fig. 11 Quantum OR operation circuit

(2,2)QMVC experiments are performed. In Sect. 4.2, (3,3)QMVC experiments are performed.

4.1 (2,2)QMVC

For a cover image, when the pixel of the cover image is white, we choose the rotation angle as $\theta = \frac{\pi}{15}$, and the probability of getting the ground state $|0\rangle$ is $\left(\cos\frac{\theta}{2}\right)^2$, as shown in Fig. 12a; when the pixel of the cover image is black, we choose the rotation angle as $\theta = \frac{14\pi}{15}$, and the probability of getting the ground state $|1\rangle$ is $\left(\sin\frac{\theta}{2}\right)^2$, as shown in Fig. 12b. The quantum states in Qiskit are read from right to left, contrary to what we set.

When constructing the quantum selection circuit, the rotation angle is set to $\theta_1 = \frac{\pi}{2.2}$, $\theta_2 = \frac{\pi}{1.5}$. The expression for the quantum state $|k\rangle$ is shown in equation 10. The quantum circuit is shown in Fig. 13a, and the probability of collapse into different ground states is shown in Fig. 13b.

When S_{ij} is selected for encryption, if the color of S_{ij} is black, the quantum circuit is shown in Fig. 14a; if the color of S_{ij} is white, the quantum circuit is shown in Fig. 14b. When C_{ij}^1 is chosen to be encrypted, if the color of C_{ij}^1 is black, the quantum

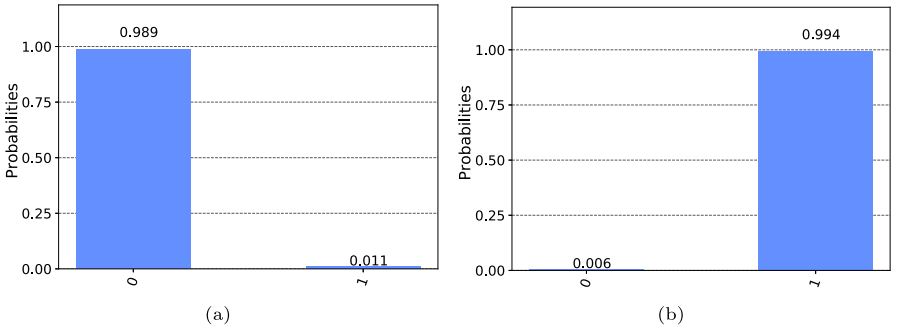


Fig. 12 **a** The probability of getting the ground state $|0\rangle$ **b** The probability of getting the ground state $|1\rangle$

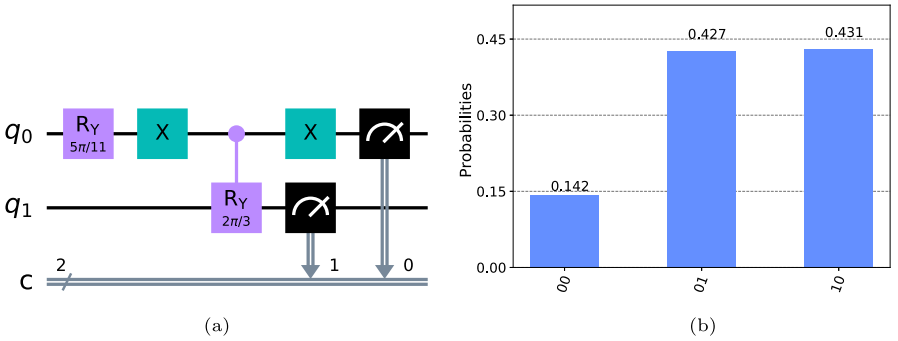


Fig. 13 **a** The quantum selection circuit for (2,2)QMVC **b** The probability of quantum selective circuit collapsing into different ground states

circuit is shown in Fig. 14c. The ground state of this circuit can be expressed as:

$$\left|C_{ij}^{1}\right\rangle_1 = \cos\frac{14\pi}{30}\cos\frac{14\pi}{30}|00\rangle + \cos\frac{14\pi}{30}\sin\frac{14\pi}{30}|01\rangle + \sin\frac{14\pi}{30}\cos\frac{14\pi}{30}|10\rangle + \sin\frac{14\pi}{30}\sin\frac{14\pi}{30}|11\rangle \tag{13}$$

If the color of C_{ij}^{1} is white, the quantum circuit is shown in Fig. 14d. The ground state of this circuit can be expressed as:

$$\left|C_{ij}^{1}\right\rangle_0 = \cos\frac{\pi}{30}\cos\frac{14\pi}{30}|00\rangle + \cos\frac{\pi}{30}\sin\frac{14\pi}{30}|01\rangle + \sin\frac{\pi}{30}\cos\frac{14\pi}{30}|10\rangle + \sin\frac{\pi}{30}\sin\frac{14\pi}{30}|11\rangle \tag{14}$$

The probabilities of the quantum circuits shown in Fig. 14 collapsing into different ground states are shown in Table 1.

Similarly, when other cover images are selected, the state of the qubit at the corresponding position is adjusted according to the pixel of the cover image, and the rest of the qubits are designed to be qubits with a high probability of $|1\rangle$.

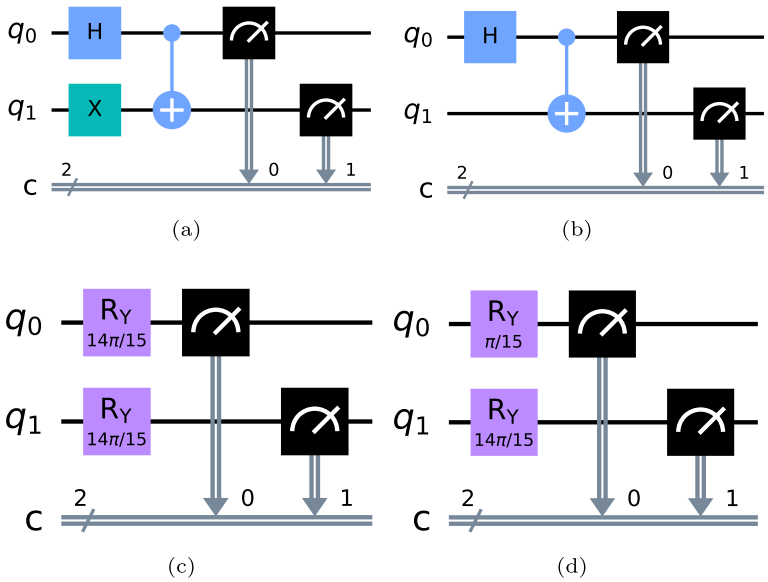


Fig. 14 (2,2)QMVC. **a** The quantum circuit when the secret image pixel is black **b** The quantum circuit when the secret image pixel is white **c** The quantum circuit when the cover image pixel is black **d** The quantum circuit when the cover image pixel is white

Table 1 The probability of secret and cover image collapsing to different ground state in (2,2) QMVC

Image	The color of pixel	The probability of collapse to a different ground state			
		$\|00\rangle$	$\|01\rangle$	$\|10\rangle$	$\|11\rangle$
Secret image	Black		0.499	0.501	
	White	0.499			0.501
Cover image	Black	0	0.011	0.011	0.978
	White	0.011	0.978	0	0.011

We used Qiskit to invoke the quantum gate and the simulation result is shown in Fig. 15, where (a) is a secret image, (b)(c) are two halftone images, (d)(e) are share images, and (f) is the recovered image.

4.2 (3,3)QMVC

When constructing the quantum selection circuit, the rotation angle is set to $\theta_1 = \frac{\pi}{1.7}$, $\theta_2 = \frac{\pi}{1.3}$, $\theta_3 = \frac{\pi}{2}$, and the quantum state is represented by the following equation:

$$|k\rangle = \cos\frac{\theta_1}{2}\cos\frac{\theta_2}{2}|00\rangle + \cos\frac{\theta_1}{2}\sin\frac{\theta_2}{2}|01\rangle + \sin\frac{\theta_1}{2}\cos\frac{\theta_3}{2}|10\rangle + \sin\frac{\theta_1}{2}\sin\frac{\theta_3}{2}|11\rangle . \tag{15}$$

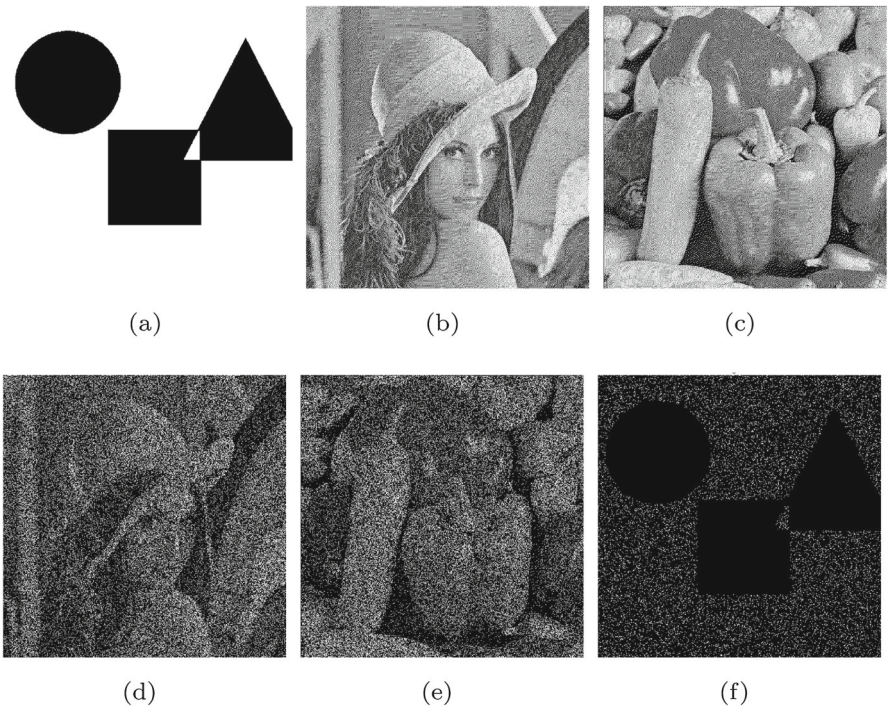


Fig. 15 (2,2)QMVC. **a** secret image **b**, **c** two halftone cover images **d**, **e** two share images **f** recovered image

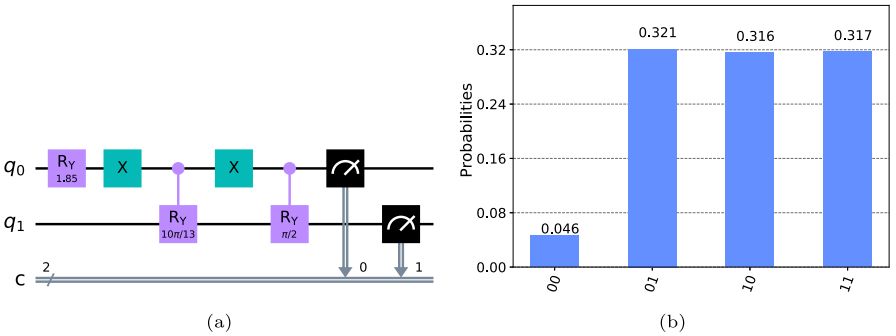


Fig. 16 **a** The quantum selection circuit for (3,3)QMVC **b** The probability of quantum selective circuits collapsing into different ground states

The quantum circuit is shown in Fig. 16a, and the probability of collapse into different ground states is shown in Fig. 16b.

When S_{ij} is selected for encryption, if the color of S_{ij} is black, the quantum circuit is shown in Fig. 17a; if the color of S_{ij} is white, the quantum circuit is shown in Fig. 17b. When C_{ij}^2 is chosen to be encrypted, if the color of C_{ij}^2 is black, the quantum circuit is shown in Fig. 17c; if the color of C_{ij}^2 is white, the quantum circuit is shown

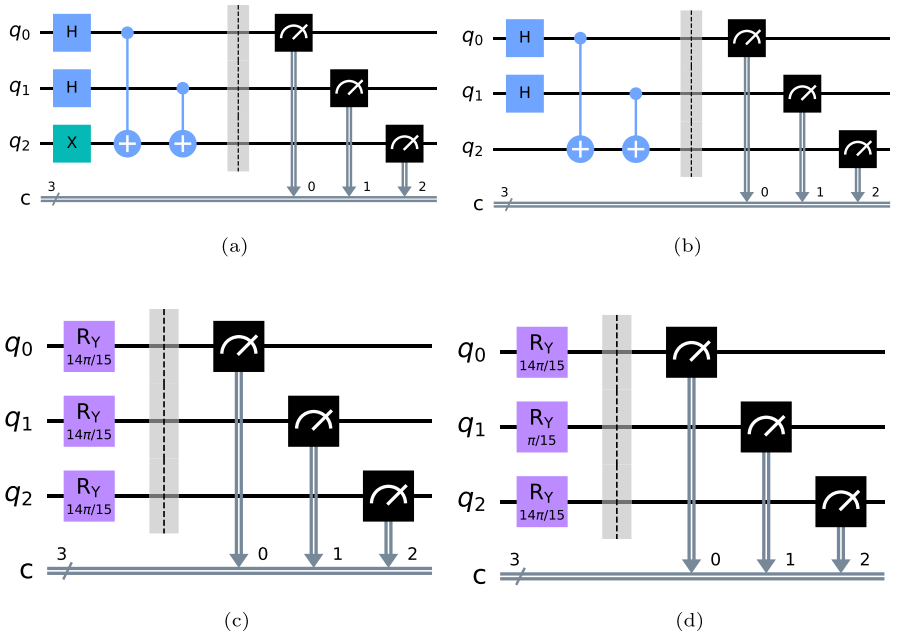


Fig. 17 (3,3)QMVC. **a** The quantum circuit when the secret image pixel is black **b** The quantum circuit when the secret image pixel is white **c** The quantum circuit when the cover image pixel is black **d** The quantum circuit when the cover image pixel is white

Table 2 The probability of secret and cover image collapsing to different ground state in (3,3)QMVC

Image	The color of pixel	The probability of collapse to a different ground state							
		$\|000\rangle$	$\|001\rangle$	$\|010\rangle$	$\|011\rangle$	$\|100\rangle$	$\|101\rangle$	$\|110\rangle$	$\|111\rangle$
Secret image	Black		0.254	0.250		0.249			0.247
	White	0.245			0.250		0.247	0.258	
Cover image	Black	0	0	0	0.019	0	0.013	0.008	0.961
	White	0	0.014	0	0	0.013	0.963	0	0.011

in Fig. 17d. The probabilities of the quantum circuits shown in Fig. 17 collapsing into different ground states are shown in Table 2. The simulation result is shown in Fig. 18, where (a) is a secret image, $(b) \sim (d)$ are three halftone images, $(e) \sim (g)$ are three share images, and (h) is the recovered image.

5 Performance evaluation

This section evaluates the performance of our proposed scheme. In Sect. 5.1, the quality of the share images is analyzed. In Sect. 5.2, the security of the QMVC scheme is analyzed. In Sect. 5.3, the scheme is compared with other quantum image encryption schemes.

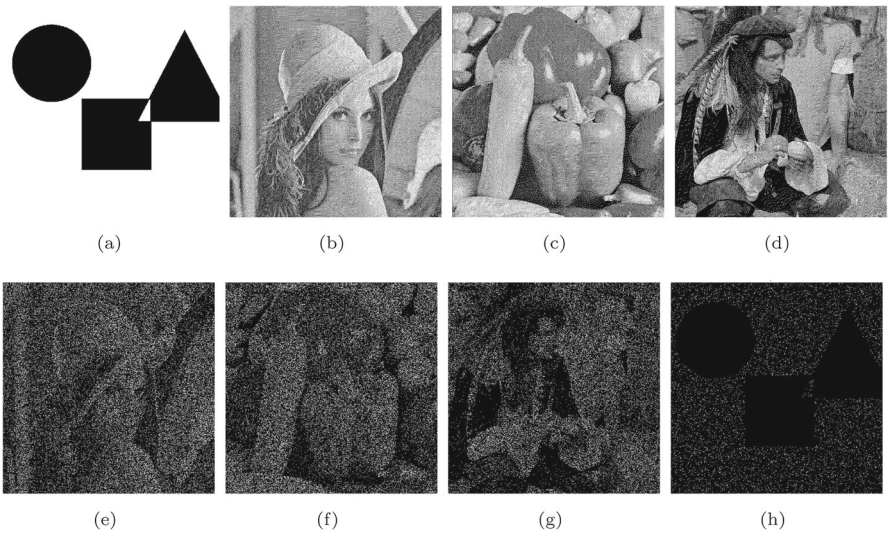


Fig. 18 (3,3)QMVC. **a** Secret image **b**–**d** three halftone cover images **e**–**g** three share images **h** recovered image

5.1 Share images quality analysis

We use PSNR and MSSIM to analyze the quality of cover images. In the experiments in Sect. 4, the PSNR of the two cover images in the (2,2)QMVC is 6.0638 and 6.1938, respectively, and the MSSIM is 0.0121 and 0.0157, respectively. The PSNR of the three cover images in the (3, 3)QMVC is 5.9293, 6.0363, and 7.6000, respectively, and the MSSIM is 0.0082, 0.0104, and 0.0217, respectively. It can be found from the same cover images in (2,2)QMVC and (3,3)QMVC that PSNR and MSSIM decrease with the increase of the number of participants. The reason is that in this scheme, encryption of secret image or cover images is selected by setting the proportion. Therefore, when the number of participants increases, the proportion allocated to each cover image will decrease, resulting in a decline in the quality of the cover image.

5.2 Security analysis

This scheme can ensure the security of the secret image. As shown in Fig. 19. Assuming that participant P_r is dishonest and has privately measured all the qubits held by himself. Since the secret information is distributed to all participants by the encryption method of the random grid, P_r cannot obtain any information of the secret image except a meaningful image. If an attacker attacks a participant and performs a quantum measurement on all his qubits. The attacker will get one meaningful image. If the attacker attacks all participants, he will get n meaningful images. Since the meaningful images look no different from normal images rather than noisy images, the attacker will not suspect that the image is a part of the secret image, thus reducing the attacker's attention and securing the secret image.

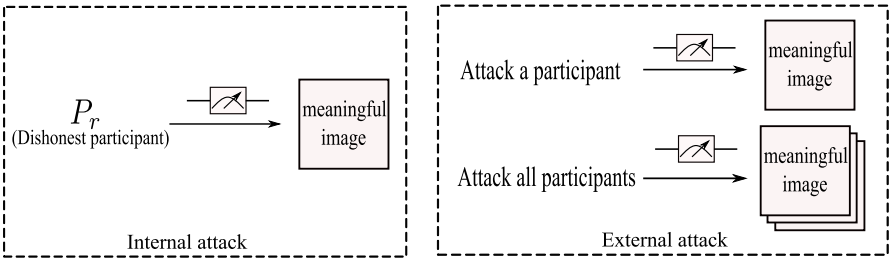


Fig. 19 Security analysis

Table 3 Comparison with existing quantum visual cryptography schemes

Scheme	Single pixel parallel processing	Pixel expansion	Security	Meaningful share image
Song et al. [20]	No	No	Yes	No
Das et al. [32]	Yes	Yes	No	No
Liu et al. [33]	Yes	No	Yes	No
Liu et al. [34]	Yes	Yes	Yes	No
Zhu et al. [35]	Yes	No	Yes	No
Our scheme	Yes	No	Yes	Yes

5.3 Comparison with existing quantum visual cryptography schemes

We compare our scheme with the proposed quantum visual cryptography schemes, as shown in Table 3. Our scheme still retains the advantage of single-pixel parallel processing. In addition, our scheme can guarantee the security of secret information in the case of dishonesty of internal participants and external attacks. Our scheme has no pixel expansion. The most important thing is that the share images in other schemes are meaningless noise images, while our scheme sets the share images as meaningful images. Our scheme can reduce the attacker's attention when subjected to external attacks.

6 Conclusion

In this paper, we propose a quantum meaningful visual cryptography scheme. In the encryption phase, we first design a quantum selection circuit. The quantum selection circuit chooses to encrypt a secret image or cover image. Then, the pixel of the secret image or the cover image is encoded as a quantum superposition state. In the decryption phase, the secret image is recovered by the quantum OR operation circuit. In our scheme, the share images are all meaningful images, further ensuring the security of the secret image. We still retain the advantage of single-pixel parallel processing in the classical VC scheme. However, our scheme still employs the strategy of qubit expansion. The secret image is a binary image and the cover image is a halftone image.

In future work, we would like to reduce the use of qubit and achieve secret and cover images to be gray or color images.

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Author Contributions All authors contributed to the study conception and design. Material preparation and data collection and analysis were performed by BY, J-SP, and H-MY. The first draft of the manuscript was written by M-YZ, and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability The datasets analyzed during the current study are available from the corresponding author on reasonable request.

Code Availability The codes used in this study are available from the corresponding author upon request.

Declarations

Conflict of interest All authors certify that they have no affiliations with or involvement in any organization or entity with any financial interest or non-financial interest in the subject matter or materials discussed in this manuscript.

Ethical approval Not applicable.

Consent to participate Not applicable.

Consent for publication Not applicable.

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